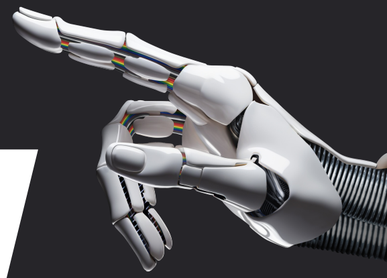
Where's My Water ?



YELLOW RUBY

SMART SOLUTION
PROJECT X
INTRODUCTION TO AI

ABOUT

YELLOW RUBY | ODOS

Position:
Project Leader, Idea Creator, Content Creator

PENGUIN
NATNARAKORN

ABOUT

YELLOW RUBY | ODOS

Position:
Video editor, Researcher about problem

TUBTIM
THANWARAT

ABOUT

YELLOW RUBY | ODOS

Position:
Researcher about feasibility

PRINCE
NONTHIKUL

ABOUT

YELLOW RUBY | ODOS

Position:
Researcher about solution, Video editor

KHAOPUN
BUNPAN

ABOUT

YELLOW RUBY | ODOS

Position:
Researcher about solution, AI agent creator,
Main actor

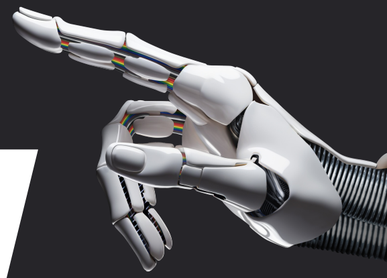
POON PHUMIPHAT

YELLOW RUBY | ODOS

ACKNOWLEDGEMENT OF COUNTRY

We respectfully acknowledge the Wurundjeri People of the Kulin Nation, who are the Traditional Owners of the land on which Swinburne's Australian campuses are located in Melbourne's east and outer-east, and pay our respect to their Elders past, present and emerging.
We are honoured to recognise our connection to Wurundjeri Country, history, culture, and spirituality through these locations, and strive to ensure that we operate in a manner that respects and honours the Elders and Ancestors of these lands.
We also respectfully acknowledge Swinburne's Aboriginal and Torres Strait Islander staff, students, alumni, partners and visitors.
We also acknowledge and respect the Traditional Owners of lands across Australia, their Elders, Ancestors, cultures, and heritage, and recognise the continuing sovereignties of all Aboriginal and Torres Strait Islander Nations.

Where's My Water ?



YELLOW RUBY

SMART SOLUTION
PROJECT X
INTRODUCTION TO AI

YELLOW RUBY | ODOS

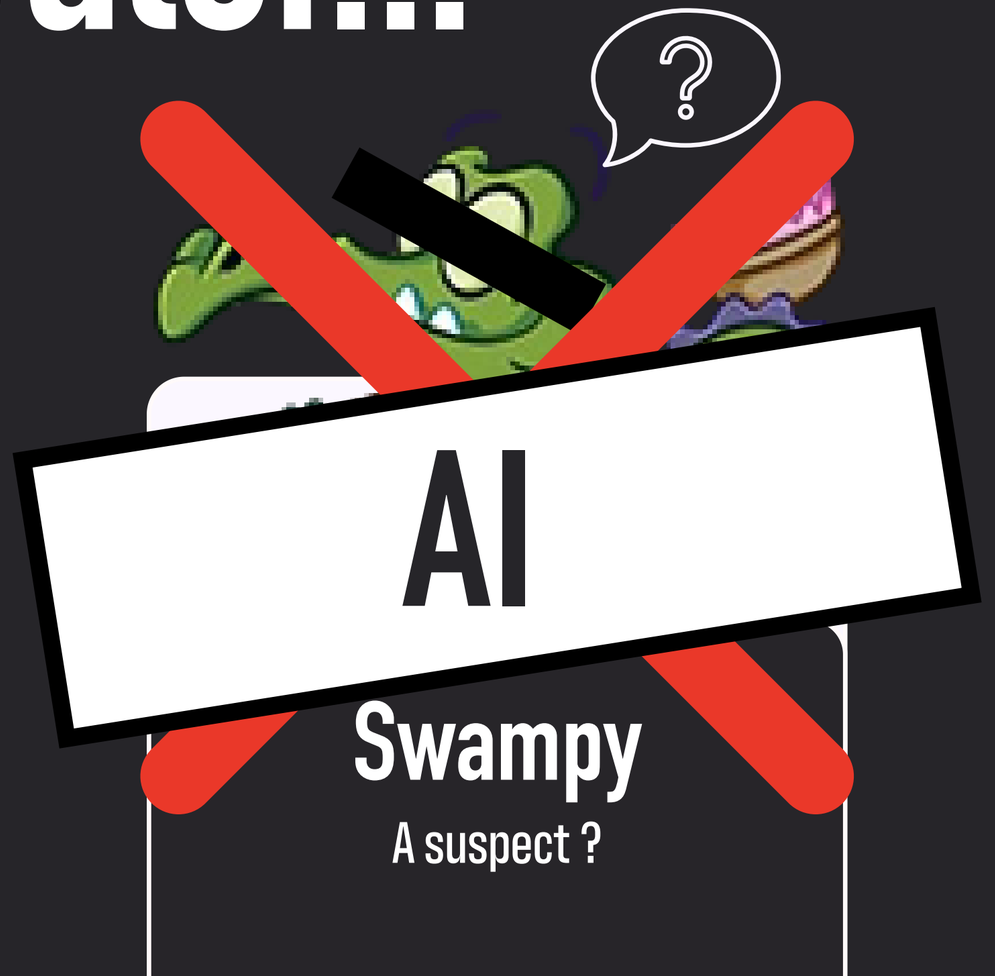
Who takes our water...

Swampy

A suspect ?

YELLOW RUBY | ODOS

Who takes our water...



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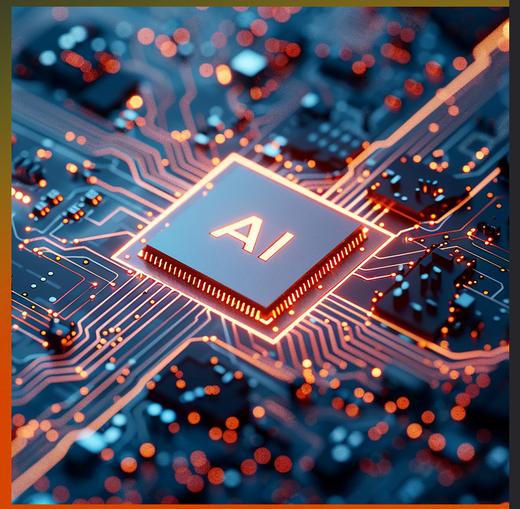
ARTIFICIAL INTELLIGENCE

WHAT IS THE COST?



NOTHING IN THIS WORLD IS FREE, AND SO DOES AI.

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WATER

Everyday life, Economic, and Essential

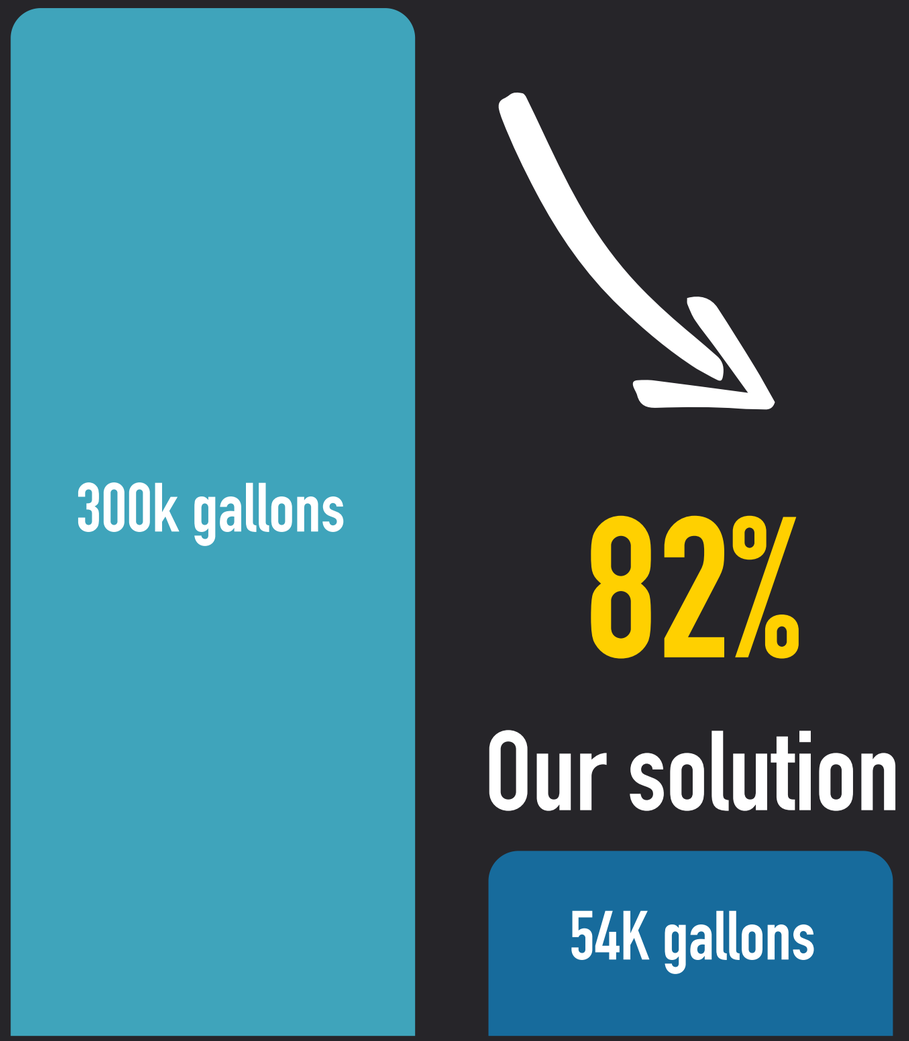
FRESH WATER INPUT. ADDITIONALLY,

YELLOW RUBY | ODOS

Impact of Direct-to-Chip Cooling

Reduction in Water Waste

- The internal dielectric fluid is continuously flow without being exposed to the outside.



Joseph W. Kane, 2025, **AI, data centers, and water** [Online], Available: https://www.brookings.edu/articles/ai-data-centers-and-water [5 May 2026].
Taryn Plumb, 2025, **Research quantifies environmental impact of liquid cooling in data centers** [Online], Available: https://www.networkworld.com/article/3979105/liquid-cooling-technologies-reducing-data-center-environmental-impact [5 May 2026].

Even though the cost is high, it's worth doing.

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REFLECTION

1

Collaboration & Coordination

2

Skills & Problem Solving

3

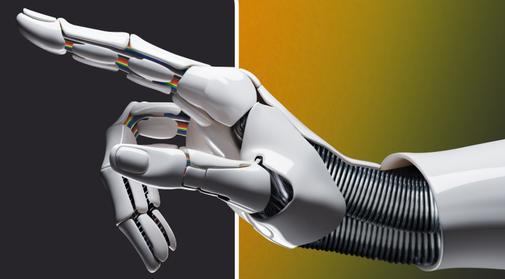
Awareness & Adaptability

1 อำเภอ 1 ทุนการศึกษา
ODOS
โอดอส ซัมเมอร์แคมป์
SWINBURNE UNIVERSITY OF TECHNOLOGY

" AI will define our future. "

The question is—will we make it sustainable?

THANK YOU

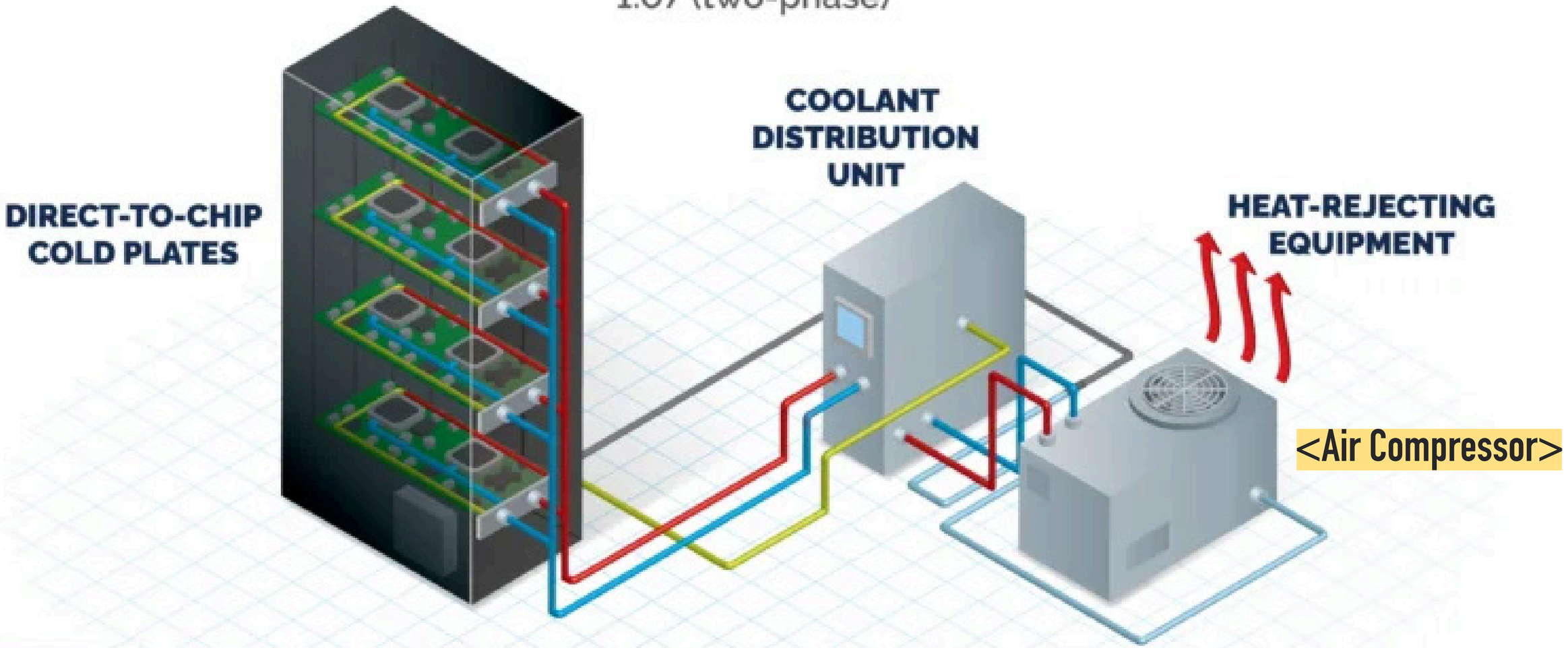


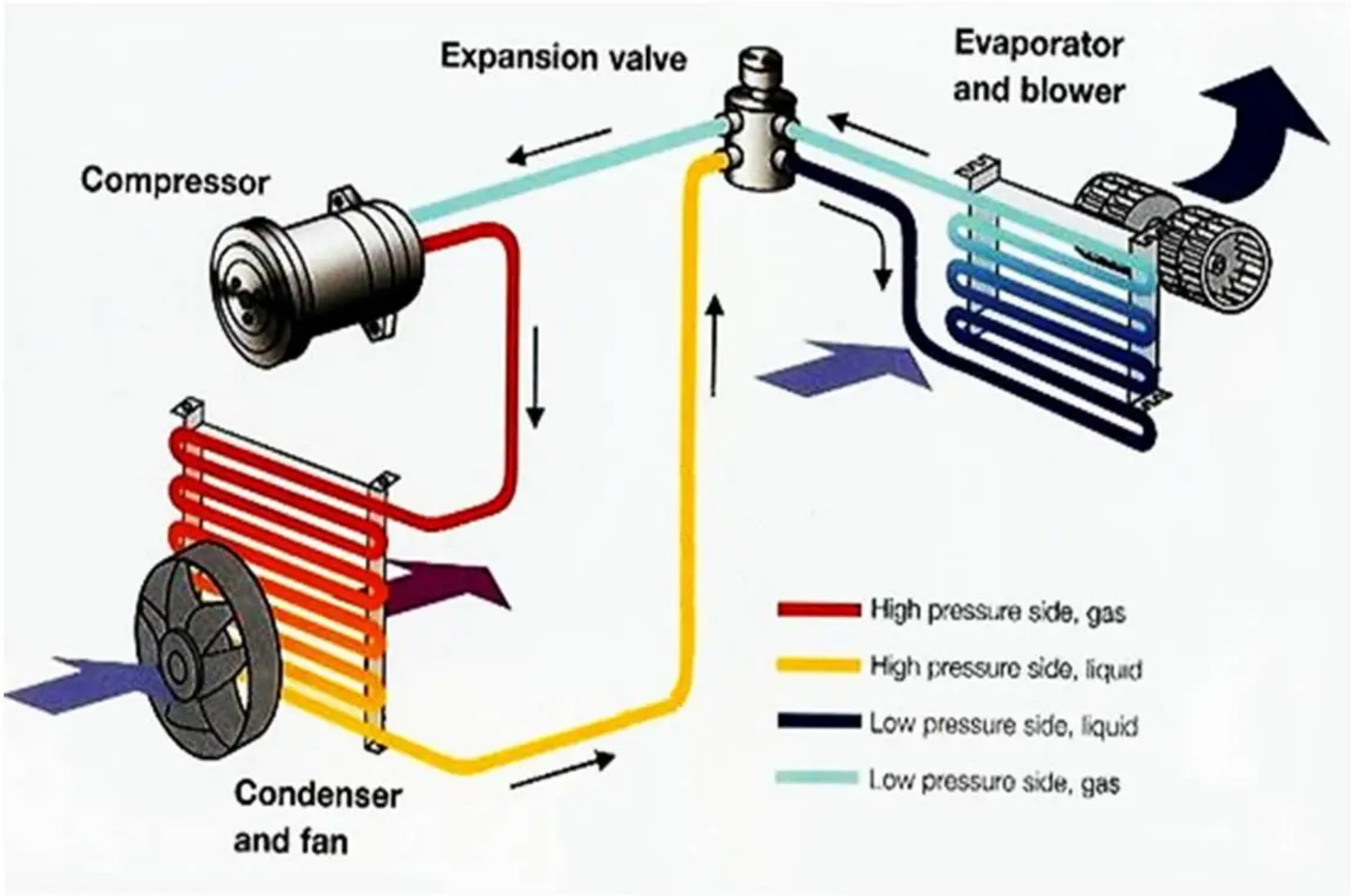
SWINBURNE UNIVERSITY OF TECHNOLOGY

1 อำเภอ 1 ทุนการศึกษา
ODOS
โอดอส ซัมเมอร์แคมป์

DIRECT-TO-CHIP COOLING

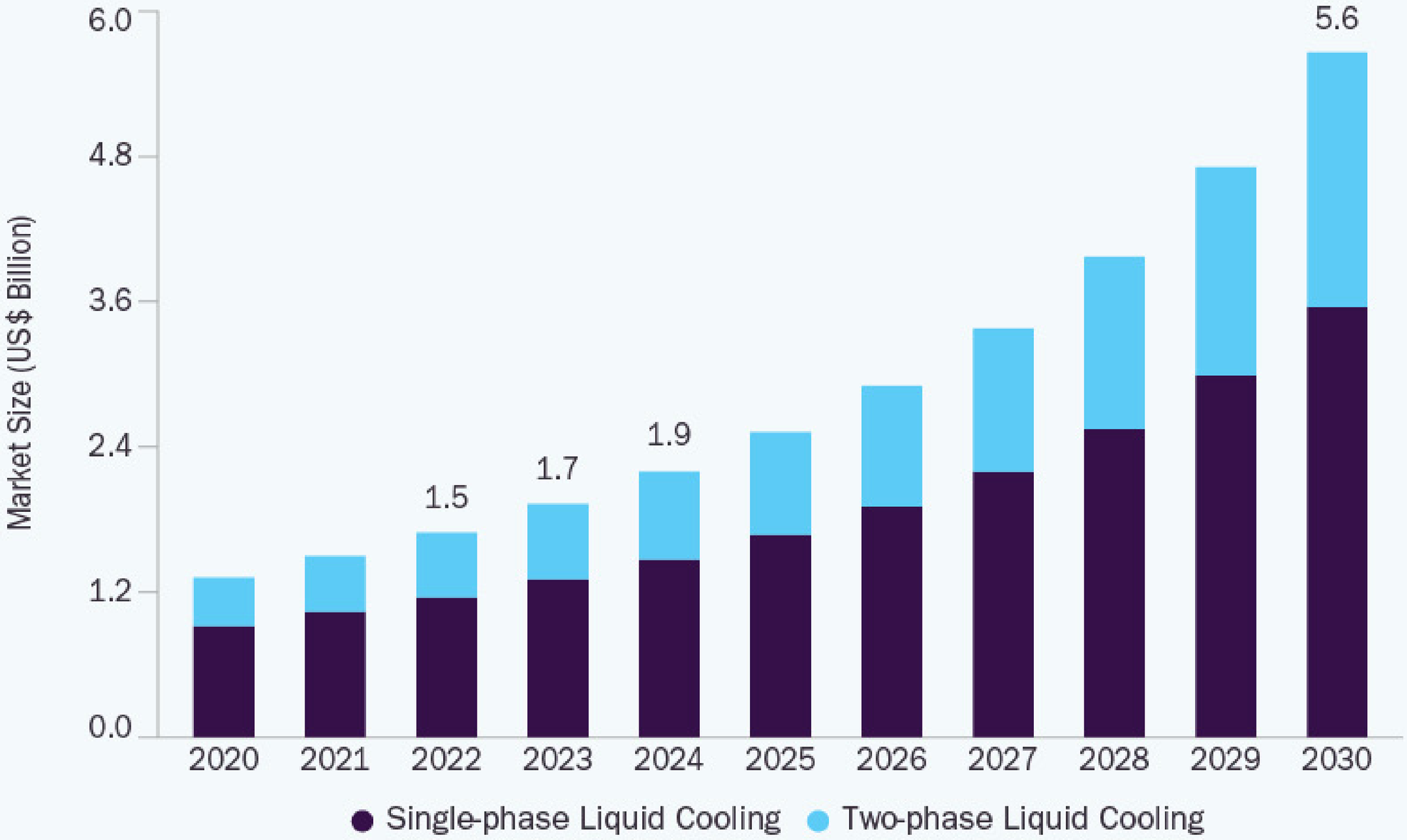
PUE = 1.13 (single-phase)
1.07 (two-phase)




Expansion valve
Evaporator
and blower
Compressor
Condenser
and fan
High pressure side, gas
High pressure side, liquid
Low pressure side, liquid
Low pressure side, gas

Direct-to-chip Liquid Cooling Market

Size, by Cooling Solution Type, 2020 - 2030 (USD Billion)



GRAND VIEW RESEARCH

19.7%

Global Market CAGR, 2025 - 2030

Source:
www.grandviewresearch.com

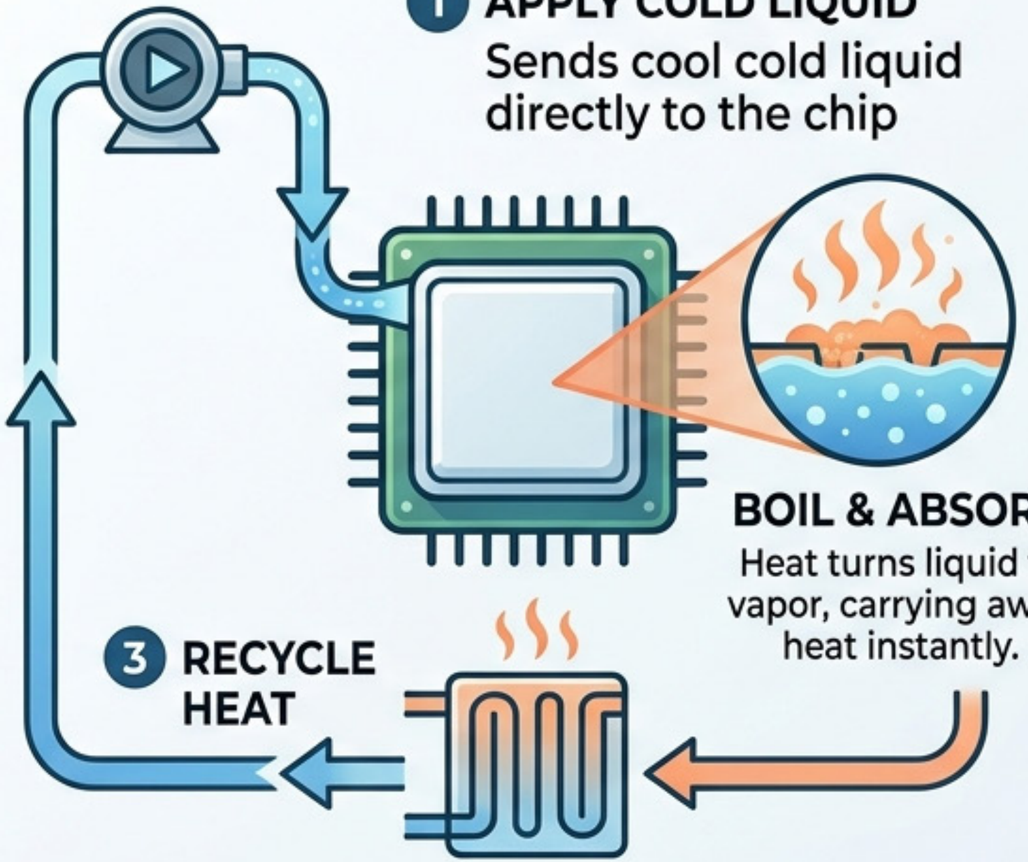
SMARTER, COOLER DATA CENTERS

THE PROBLEM

Air Cooling Can't Keep Up.

New Computers Generate Too Much Heat.

THE TWO-PHASE SOLUTION



SEALED CLOSED LOOP

KEY BENEFITS

ENERGY SAVINGS
Uses Less Power for Cooling

WATER SAVINGS
Minimal Water Loss in Closed System

SPACE EFFICIENCY
More Power in Less Space

LOGO

BUILDING THE FINANCIAL CASE FOR TWO-PHASE COOLING

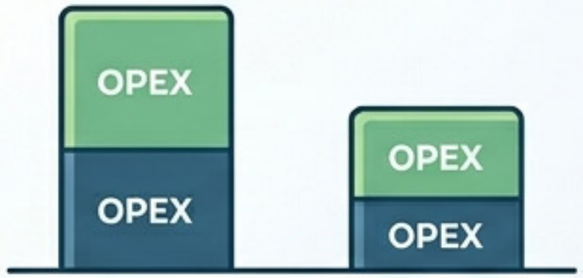
1. GLOBAL MARKET OPPORTUNITY

AI/HPC

Hyperscalers

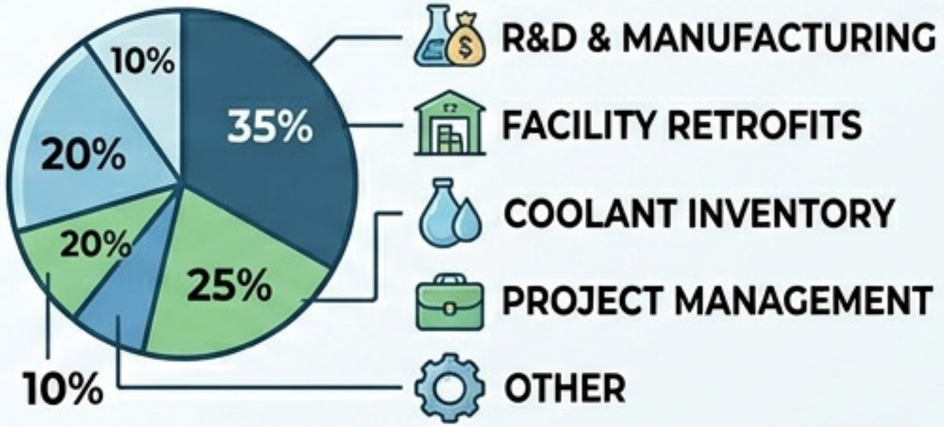
TAM (2030): **$15 BILLION.**
CAGR (2024-2030): **12%.**

2. TOTAL COST OF OWNERSHIP (TCO) COMPARISON (5-YEAR)



Lower **OPEX** offsets higher per-rack **CAPEX.**

3. ESTIMATED INITIAL INVESTMENT BREAKDOWN



4. MONTHLY OPERATIONAL SAVINGS (FACILITY LEVEL)

A

POWER SAVINGS
-40%
Electricity

B

SPACE EFFICIENCY
6x Density
- Less Real Estate Cost

C

LOWER MAINTENANCE

5. PROJECTED FINANCIALS & PAYBACK

IRR 30%

Payback Period:
24 MONTHS.

PILOT LIVE
(Year 1)

BREAK-EVEN
(Year 2)

POSITIVE NPV
(Year 3+)

PHASE_COOL

INVESTOR PITCH 2024